

CSCI 611 - Assignment 4

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1 Implementation

This section summarizes the main TODOs completed in the notebook.

1.1 Feature Extraction

The pretrained VGG network was used to extract activations from selected layers. The chosen layers were:

```
layers = {  
    '0': 'conv1_1',  
    '5': 'conv2_1',  
    '10': 'conv3_1',  
    '19': 'conv4_1',  
    '21': 'conv4_2',  
    '28': 'conv5_1'  
}
```

Here, conv4_2 was used for content, while the others contributed to style.

1.2 Gram Matrix Implementation

One important TODO was implementing the Gram matrix:

```
def gram_matrix(tensor):  
    batch_size, depth, height, width = tensor.size()  
    features = tensor.view(batch_size, depth, height * width)  
    gram = torch.bmm(features, features.transpose(1, 2)) / (depth * height  
        * width)  
    return gram
```

This normalization helped stabilize the style loss across layers.

1.3 Loss Computation

The content loss was computed as:

```
content_loss = torch.mean(  
    (target_features[content_layer] - content_features[content_layer]) **  
    2  
)
```

The style loss was computed across all selected style layers:

```

style_loss = 0
# iterate through each style layer and add to the style loss
for layer in style_weights:
    # get the "target" style representation for the layer
    target_feature = target_features[layer]
    _, d, h, w = target_feature.shape

    target_gram = gram_matrix(target_feature)
    style_gram = style_grams[layer]
    layer_style_loss = style_weights[layer] * torch.mean((target_gram
        - style_gram)**2)

    # add to the style loss
    style_loss += layer_style_loss

```

The total loss was:

```
total_loss = content_weight * content_loss + style_weight * style_loss
```

1.4 Optimization

The target image was optimized with Adam:

```
optimizer = optim.Adam([target], lr=0.03)
```

After each update, the target image was clamped to keep pixel values in a valid range:

```

with torch.no_grad():
    target.clamp_(0, 1)

```

2 Experimental Setup

In Part 2, I explored the effect of different hyperparameter values. The main parameters varied were:

- style weight β ,
- content weight α ,
- learning rate,
- number of optimization steps,
- relative style layer weights.

2.1 Baseline Setting

The baseline configuration used:

```

style_weights = {
    'conv1_1': 1.0,
    'conv2_1': 0.8,
    'conv3_1': 0.5,

```

```

    'conv4_1': 0.3,
    'conv5_1': 0.1
}

content_weight = 1
style_weight = 1e7
steps = 5000
learning_rate = 0.03

```

3 Results

This section presents visual and qualitative comparisons across experiments.

3.1 Experiment 1: Baseline

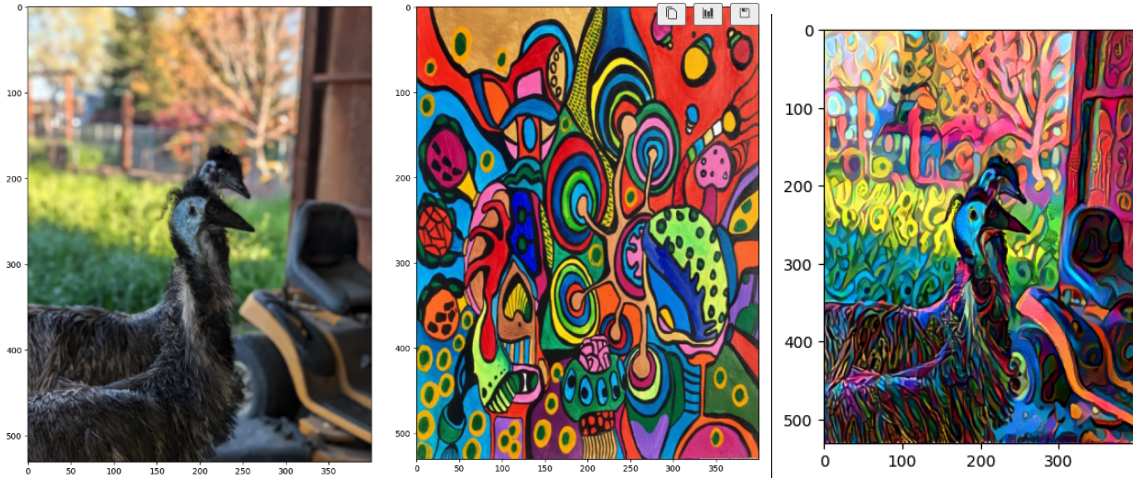


Figure 1: Left: content image, middle: style image, right: stylized result using baseline settings.

Observation: The baseline configuration produces a balanced result where both the content structure and stylistic features are preserved. The image shows noticeable stylization while still maintaining recognizable object shapes.

3.2 Experiment 2: Lower Style Weight

```

style_weights = {
    'conv1_1': 1.0,
    'conv2_1': 0.8,
    'conv3_1': 0.5,
    'conv4_1': 0.3,
    'conv5_1': 0.1
}

content_weight = 1
style_weight = 1e3
steps = 5000

```

```
learning_rate = 0.03
```

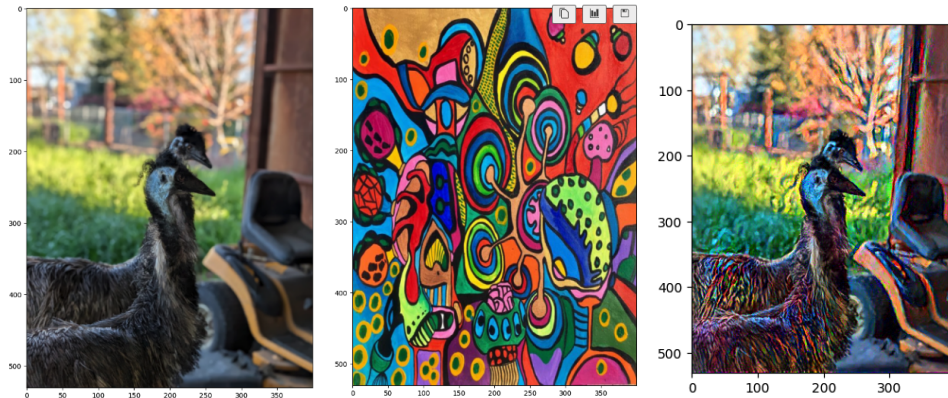


Figure 2: Left: content image, middle: style image, right: stylized result using lower style weight settings.

Observation: With a lower style weight, the output preserved more of the original content structure but showed weaker stylization.

3.3 Experiment 3: Higher Style Weight

```
style_weights = {  
    'conv1_1': 1.0,  
    'conv2_1': 0.8,  
    'conv3_1': 0.5,  
    'conv4_1': 0.3,  
    'conv5_1': 0.1}  
content_weight = 1  
style_weight = 1e25  
steps = 5000  
learning_rate = 0.03
```

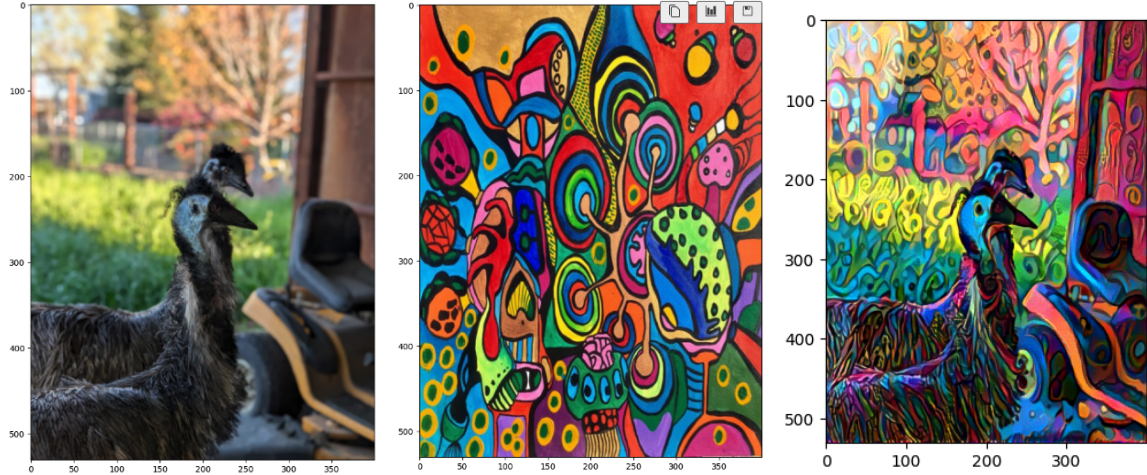


Figure 3: Left: content image, middle: style image, right: stylized result using higher style weight settings.

Observation: Increasing the style weight significantly caused the style loss to dominate the optimization, leading to strong texture transfer and color distortion. While the artistic effect became more pronounced, the original content structure became less recognizable.

3.4 Experiment 4: Different Layer Weights

```
style_weights = {
    'conv1_1': 10.0,
    'conv2_1': 5.0,
    'conv3_1': 1.0,
    'conv4_1': 0.0,
    'conv5_1': 0.0
}

content_weight = 1
style_weight = 1e9
steps = 5000
learning_rate = 0.03
```

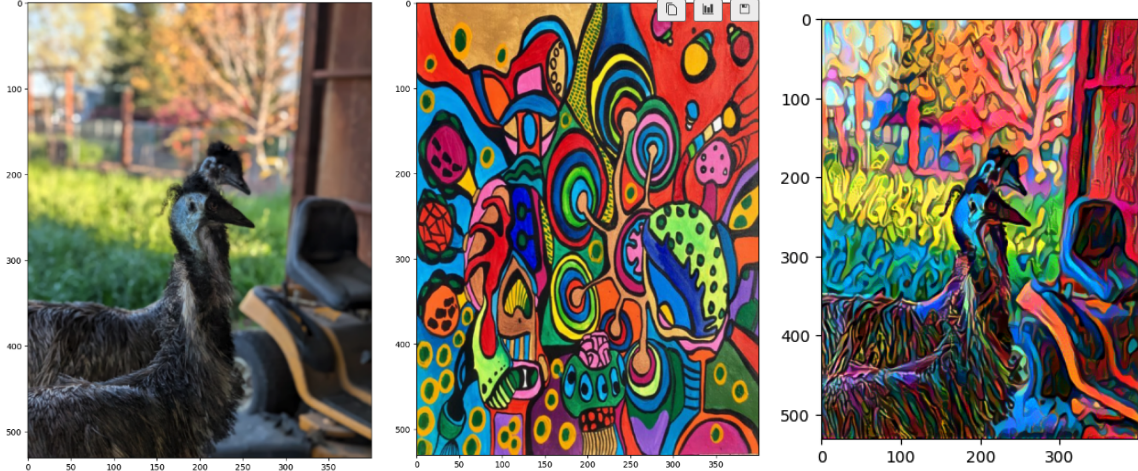


Figure 4: Left: content image, middle: style image, right: stylized result using higher weight for earlier layers.

Observation: Increasing the weights of earlier convolutional layers (conv1_1 and conv2_1) resulted in a noticeable increase in fine-grained textures and color fragmentation. Compared to the baseline configuration, the output image exhibited more local variations and less coherent structure. However, the overall visual difference remained relatively subtle because the baseline already used a high global style weight, meaning the image was already strongly stylized. Thus, adjusting layer weights primarily changed the distribution of stylistic features rather than the overall strength of stylization.

4 Comparison Table

Experiment	Style Weight	LR	Steps	Main Observation
Baseline	10^7	0.03	5000	Balanced content and style
Low Style	10^3	0.03	5000	Weak stylization
High Style	10^{25}	0.03	5000	Strong style, less content fidelity
Layer Emphasis	10^9	0.03	5000	More visible texture artifacts

Table 1: Comparison of selected hyperparameter settings.

5 Discussion

The experiments showed that the most important parameter affecting the appearance of the result was the ratio between content and style loss. When the style weight was too small, the target image barely changed. When the style weight was too large, the output became heavily stylized and sometimes lost recognizable content structure.

The choice of style image also mattered. Images with strong textures and brushstrokes produced more dramatic results, while smoother paintings produced subtler stylistic changes. The Mona Lisa, for example, can lead to a relatively subtle transfer because its style is less textured than paintings such as those by Van Gogh.

The selected style layers also influenced the appearance. Earlier layers contributed more local texture, while deeper layers contributed broader structural patterns.